HEDF PROJECT SUMMARY SHEET

COMMISSION ON HIGHER EDUCATION

Proposed for Calendar Year **2022**

1. **Proponent Code** — *Project Code to be filled up by HEDF staff for new projects*

2. **Project Title**: **Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay**

3. **Program/ Project Description**: This study would like to assess the mangrove forest structure in San Miguel Bay and its potential implications to fishery and other ecosystem services. In addition, the willingness-to-pay among the fisherfolks and coastal communities will be determined. Furthermore, this study will also look into interventions and startegies appropriate for the communities in order to properly manage and conserve the mangrove forests in San Miguel Bay. This study will employ mixed methods (quantitative and qualitative). The mangrove forest structure will be determined by establishing sampling plots for mangrove forest mensuration. The implication of mangrove forest structure to fishery and other ecosystem services, the researchers will implement purposive sampling, targetting fisher folks and coastal communities, in order to determine qualitative and quantitative data. In case of determining willingness-to-pay amonh the targetted stakeholders, mixed methods of Focus Group Discussion, Household Survey, and Individual interview will be undertaken, employing purposive sampling. The same method will be used to determine the appropriate strategies for mangrove conservation, by determining the perception of communities towards mangrove forests and conservation practices.

4. **Proponent Name**: **DIVINA P. CHICA** — Signature

5. **Proponent Org/Div**: **PARTIDO STATE UNIVERSITY - TINAMBAC CAMPUS**

6. **Contact Information**: **Contact Number/Details:** 09366571568/ divina.chica@parsu.edu.ph/ (054) 871-2090 / op@parsu.edu.ph
Address: Tinambac, Camarines Sur

7. **Project Dates and Duration**: From ____ to ____ Duration **24** months

8. **Requested Support**

Funding Amount: For the year N/A — For the Entire Project **Php4,500,000.00**

Other Request

9. **HEDF Priority Areas**

10. **Project Justification**: The proposed research and development project is significantly relevant to the the efforts being exerted towards achieving sustainable development. It is a study related to mangrove forest and mangrove ecosystem in general, which is designed to gather relevant sets of information, which are urgently needed; not only to San Miguel Bay but also to the total national mangrove production (or depletion) in the country. Moreover, this study can provide insights once it successfully produce the deliverables comprising the following: (1) Vegetation cover assessment of mangrove forest, which includes a vegetation map, and stand composition/diversity; (2) Factors related to stakeholders and decision making in SMB particularly those that explain mangrove forest planning, utilization, and management; (3) List of stakeholders and the factors/ motivational drivers affecting their behavior towards the utilization and management of mangrove forest, and (4) lastly, the use of CVM, Contingent Valuation Method that will determine the PES or payment for ecosystem services using WTP or the stakeholders' willingness to pay technique. Furthermore, the outputs of this study may be used as preliminary data and reference to feasibility assessment of Payment for Ecosystem Services in San Miguel Bay and even in nearby communities with similar circumstances.

ENDORSEMENT

11. **Program / Project Deliverables**:
The following are expected deliverables of the project:
1. Five Publications:
   (a) Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay
   (b) Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay
   (c) Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities of San Miguel Bay
   (d) Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay.
   (e) Willingness to Pay for the Conservation of Mangrove Forest among Fisherfolks and Coastal Communities of San Miguel Bay
2. Manual for Publication: Mangrove Species and Conservation Management
3. Consolidated Terminal /Final Report for the project
4. Financial Report / Report of Fund Disbursement
5. Accomplished Property Inventory Form

12.

	Project Milestones	Date	Output
1.	Development of preliminary questionnaires for four project components (components 2-5)	Month 1-3	Four sets of preliminary questionnaires for four project components i.e., (a) Fishery Productivity in San Miguel Bay (SMB), (b) INventory of Ecosystem Services derived from SMB, (c) Socio-psychological factors affecting community utilization and participation in mangrove conservation and management, and (d) Willingness to Pay among coastal communities of SMB for the conservation of mangrove ecosystem
2.	Pretesting of the developed questionnaires	Month 4-5	Four sets of Pre-tested Questionnaires
3.	Validation of the developed Food Frequency Questionnaire	Month 5-6	Validated questionnaires; Scenario definition of the contingent valuation study is finalized

4.	Validation of the developed questionnaires		Month 6-8				Updated socio-demographic data of involved communities; and final sampling plan for the the project components
5.	Secondary data collection		Month 7-8				Final sampling plan for field data gathering for the project component: vegetative assessment of mangrove stands of San Miguel Bay
6	Estimation of mangrove forest cover using remote sensing		Month 8-12				Publication 1: Community Structure and Biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay
7	Survey on factors affecting community utilization and participation in mangrove conservation and management		Month 8-12				Publication 2: Socio-Psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay
Project Milestones		**Date**					**Output**
8	Inventory of community-perceived ecosystem services derived from Mangrove ecosystems of San Miguel Bay (SMB)		Month 13-18				Publication 3: Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities of San Miguel Bay
9	Contingent valuation study on Mangrove Forest Ecosystem		Month 8-19				Publication 5: Willingness to Pay for the Conservation of Mangrove Forest among Fisherfolks and Coastal Communities of San Miguel Bay
10	Estimation of fishery resource stocks of San Miguel Bay using rapid resource assessment and catch per Unit Effort methods		Month 8-21				Publication 4: Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay.
11	Compilation of findings from all project components		Month 13-15; 18-21				Research-based data on the status of mangrove ecosystem services of San Miguel Bay; and basis for IEC campaign and manual
12	Information, education, and communication (IEC) campaign on Mangrove Ecosystems of SMB		Month 13-15; 18-21				Locally-adapted IEC pamphlets and other materials and webinar series on the management and conservation of Mangrove Ecosystems of San Miguel Bay
13	Drafting of report, Terminal Report		Month 18-24				Manual production; and research and development program completion

13.

Activity Chart	**Mo.1**	**Mo.2**	**Mo.3**	**Mo.4**	**Mo.5**	**Mo.6**	**Mo.7**	**Mo.8**	**Mo.9**	**Mo.10**	**Mo.11**	**Mo.12**
Year 1												
Development of questionnaires												
Pretesting of the developed questionnaires												
Validation of the developed questionnaires												
Secondary data collection												
Estimating mangrove forest cover using remote sensing												
Vegetative analysis of Mangrove stands of San Miguel Bay												
Survey on factors affecting community utilization and participation in mangrove conservation and management												
Contingent valuation study on Mangrove Forest Ecosystem												

	Estimation of fishery resource stocks of San Miguel Bay using rapid resource assessment and catch per Unit Effort methods												
Activity Chart		**Mo.13**	**Mo.14**	**Mo.15**	**Mo.16**	**Mo.17**	**Mo.18**	**Mo.19**	**Mo.20**	**Mo.21**	**Mo.22**	**Mo.23**	**Mo.24**
Year 2													
	Inventory of community-perceived ecosystem services derived from Mangrove ecosystems of San Miguel Bay												
	Contingent valuation study on Mangrove Forest Ecosystem												
	Estimation of fishery resource stocks of San Miguel Bay using rapid resource assessment and catch per Unit Effort methods												
	Compilation of findings from all project components												
	Information, education, and communication (IEC) campaign on Mangrove Ecosystems of San Miguel Bay												
	Drafting of report, Terminal Report												

Financial Timelines	**Mo.1**	**Mo.2**	**Mo.3**	**Mo.4**	**Mo.5**	**Mo.6**	**Mo.7**	**Mo.8**	**Mo.9**	**Mo.10**	**Mo.11**	**Mo.12**	**TOTAL (PhP)**
Travel/Transportation Expenses		45,250.00				45,250.00	45,250.00	45,250.00	45,250.00	45,250.00	37,225.00		**308,725.00**
Supplies and materials						249,974.00							**249,974.00**
Representation Expenses		20,500.00				73,000.00	7,000.00	7,000.00	7,000.00	7,000.00	18,000.00		**139,500.00**
Honoraria	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	**555,600.00**
Contract of Service	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	**489,084.00**
Enumerators' Fee												59,700.00	**59,700.00**
Equipment						293,000.00							**293,000.00**

Financial Timelines	**Mo.13**	**Mo.14**	**Mo.15**	**Mo.16**	**Mo.17**	**Mo.18**	**Mo.19**	**Mo.20**	**Mo.21**	**Mo.22**	**Mo.23**	**Mo.24**	**TOTAL (PhP)**
Travel/ Transportation Expenses	45,250.00		45,250.00			45,250.00			45,250.00	37,225.00		45,250.00	**263,475.00**
Supplies and materials													**0.00**
Representation Expenses										18,000.00		73,000.00	**91,000.00**
Honoraria	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	46,300.00	**555,600.00**
Contract of Service	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	40,757.00	**489,084.00**
Enumerators' Fee						57,600.00	99,500.00		168,000.00				**325,100.00**
Equipment	42,000.00												

Publication, Printing and Binding Expenses										146,666.00	146,667.00	146,667.00	440,000.00
TOTAL	**261,364.00**	**239,864.00**	**219,364.00**	**174,114.00**	**174,114.00**	**938,188.00**	**325,864.00**	**226,364.00**	**439,614.00**	**428,255.00**	**376,006.00**	**498,731.00**	**4,259,842.00**
Administrative Cost (5%)													**198,158.00**
TOTAL													**4,500,000.00**

15 REVIEWED BY:

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